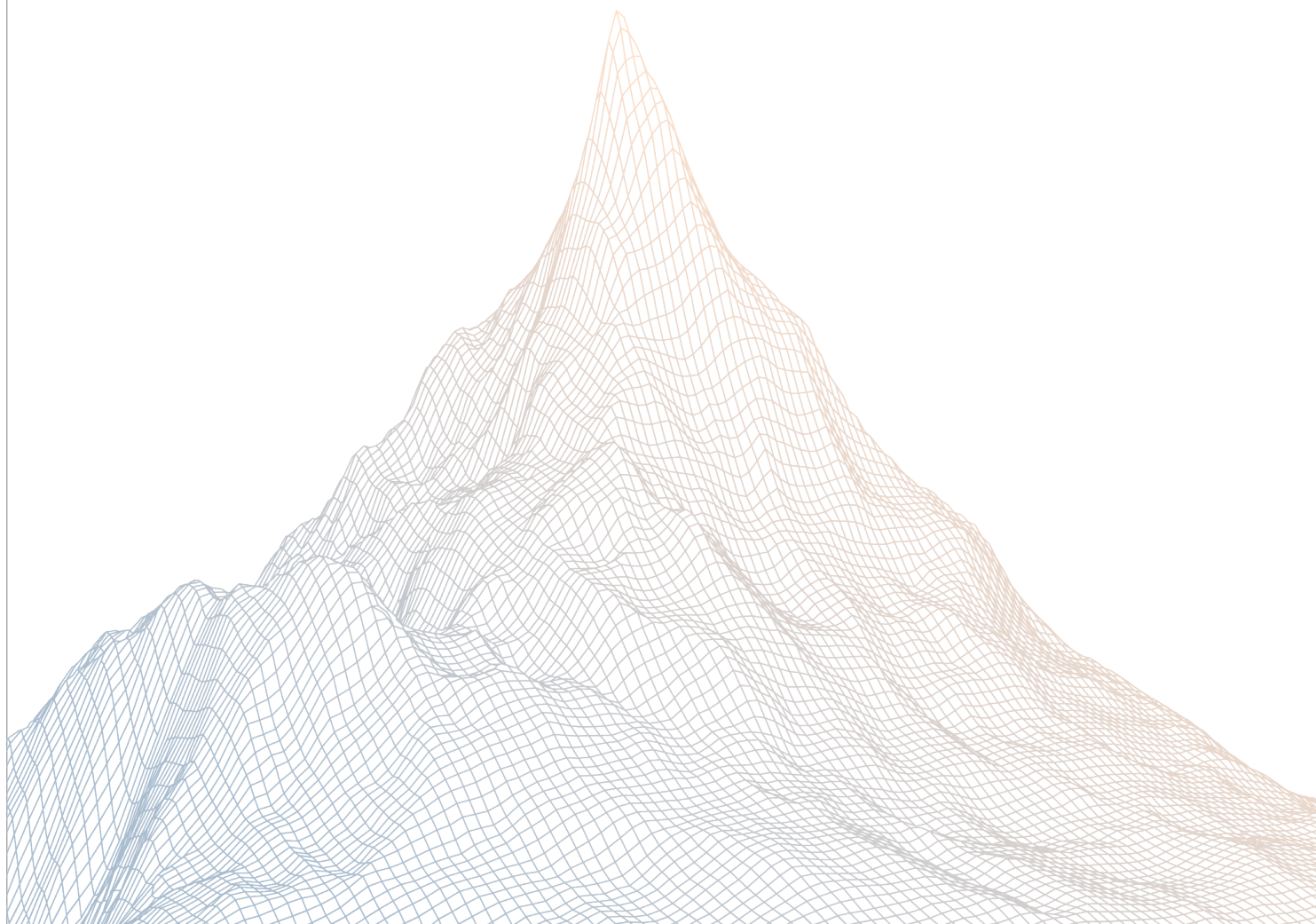


Meteora

Smart Contract Security Assessment

VERSION 1.1



AUDIT DATES: June 19th to June 20th, 2025
AUDITED BY: Mario Ponder
peakbolt

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Introduction

1.1 About Zenith

Zenith assembles auditors with proven track records: finding critical vulnerabilities in public audit competitions.

Our audits are carried out by a curated team of the industry's top-performing security researchers, selected for your specific codebase, security needs, and budget.

Learn more about us at <https://zenith.security>.

1.2 Disclaimer

This report reflects an analysis conducted within a defined scope and time frame, based on provided materials and documentation. It does not encompass all possible vulnerabilities and should not be considered exhaustive.

The review and accompanying report are presented on an "as-is" and "as-available" basis, without any express or implied warranties.

Furthermore, this report neither endorses any specific project or team nor assures the complete security of the project.

1.3 Risk Classification

SEVERITY LEVEL	IMPACT: HIGH	IMPACT: MEDIUM	IMPACT: LOW
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

2

Executive Summary

2.1 About Meteora DAMM V2

Our mission is to build the most secure, sustainable and composable liquidity layer for all of Solana and DeFi.

By using Meteora's DLMM and Dynamic AMM Pools, liquidity providers can earn the best fees and yield on their capital.

This would help transform Solana into the ultimate trading hub for mainstream users in crypto by driving sustainable, long-term liquidity to the platform. Join us at Meteora to shape Solana's future as the go-to destination for all crypto participants.

2.2 Scope

The engagement involved a review of the following targets:

Target	damm-v2
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Repository	https://github.com/MeteoraAg/damm-v2
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Commit Hash	f038ba6d13f99948270ff94a315874a1df244fe7
--------------------	--

Files	Changes in the PR-52
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2.3 Audit Timeline

June 19, 2025	Audit start
June 20, 2025	Audit end
June 26, 2025	Report published

2.4 Issues Found

SEVERITY	COUNT
Critical Risk	0
High Risk	0
Medium Risk	0
Low Risk	0
Informational	1
Total Issues	1

3

Findings Summary

ID	Description	Status
I-1	Admin cannot initialize or update reward at index zero	Resolved

4

Findings

4.1 Informational

A total of 1 informational findings were identified.

[I-1] Admin cannot initialize or update reward at index zero

SEVERITY: Informational

IMPACT: Informational

STATUS: Resolved

LIKELIHOOD: Low

Target

- [programs/cp-amm/src/state/pool.rs#L744-L758](#)
- [programs/cp-amm/src/instructions/admin/ix_initialize_reward.rs#L62](#)
- [programs/cp-amm/src/instructions/admin/ix_update_reward_duration.rs#L44](#)
- [programs/cp-amm/src/instructions/admin/ix_update_reward_funder.rs#L25](#)

Description

When a pool has a non-default creator set, the admin is unable to initialize the reward at `reward_index = 0`, nor update its reward duration or funder. This is due to the logic in `Pool::validate_authority_to_edit_reward`:

```
pub fn validate_authority_to_edit_reward(
    &self,
    reward_index: usize,
    signer: Pubkey,
) → Result<()> {
    // legacy pools would have creator's pubkey as Pubkey::default()
    if reward_index == 0 && self.creator.ne(&Pubkey::default()) {
        // only pool creator is allowed to initialize reward with index 0
        require!(self.creator == signer, PoolError::InvalidPoolCreator);
    } else {
        require!(assert_eq_admin(signer), PoolError::InvalidAdmin);
    }
    Ok(())
}
```

This function is called by the `validate` function in the following instructions:

- `initialize_reward`
- `update_reward_duration`
- `update_reward_funder`

Recommendations

It is recommended to consider updating the logic in `validate_authority_to_edit_reward` to allow admins to initialize or update reward at index zero, even when a pool creator is set, if that is the intended behavior. If only the creator should have this authority, it is beneficial to clarify this restriction in the documentation and comments.

Meteora: Resolved with [PR-76](#)

Zenith: Verified.